

Can information theory provide a scientific basis for machine learning applications related to space weather?

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Information theory can be integrated with machine learning to achieve greater insights and performance

Information theory can help untangle, prune, and select the input parameters for ML-based modeling by identifying and ranking the relevant parameters that actually transfer information to the model output parameters.

As a result, it can help reduce the dimensionality of the networks, shorten the training time, and improve the model accuracies.



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contributed equally to this work.

Key Points:

- Evidence of direct causality from relativistic into ultra-relativistic electrons, compatible with local acceleration in the outer belt.
- Detection of information transfer unveils the mechanisms of energy transfer in radiation belts, important for space weather forecasting.
- Information flow formulation of causality has a great potential for space physics discoveries.

Supporting Information:
Supporting Information may be found in the online version of this article.

Causal Inference in the Outer Radiation Belt: Evidence for Local Acceleration

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Abstract Currently, there is no clear understanding of the comprehensive set of variables that controls fluxes of relativistic electrons within the outer radiation belt. Herein, the methodology based on causal inference is applied for identification of factors that control fluxes of relativistic electrons in the outer belt. The patterns of interactions between the solar wind, geomagnetic activity and belt electrons have been investigated. We found a significant information transfer from solar wind, geomagnetic activity and fluxes of very low energy electrons (54 keV), into fluxes of relativistic (470 keV) and ultra-relativistic (2.23 MeV) electrons. We present evidence of a direct causal relationship from relativistic into ultra-relativistic electrons, which points to a local acceleration mechanism for electrons energization. It is demonstrated that the observed information transfer from low energy electrons at 54 keV into energetic electrons at 470 keV is due to the presence of common external drivers such as substorm activity.



Space Weather^{*}

RESEARCH ARTICLE
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Key Points:

- An empirical model to predict state of radiation belt relativistic electrons is developed.
- The model prediction efficiency increases with increasing L^* with a PE > 0.6 at $L^* > 5$.
- The model can potentially complement a class of empirical models that input observations from low earth orbit.

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Modeling Radiation Belt Electrons With Information Theory Informed Neural Networks

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Abstract An empirical model of radiation belt relativistic electrons ($\mu = 560\text{--}875\text{ MeV G}^{-1}$ and $I = 0.088\text{--}0.14 R_g\text{ G}^{0.2}$) with average energy $\sim 1.3\text{ MeV}$ is developed. The model inputs solar wind parameters (velocity, density, interplanetary magnetic field (IMF) (B_z and B_y), magnetospheric state parameters (SYM-H and AL), and L^* . The model outputs the radiation belt electron phase space density (PSD). The model is operational from $L^* = 3$ to 6.5. The model is constructed with neural networks assisted by information theory. Information theory is used to select the most effective and relevant solar wind and magnetospheric input parameters plus their lag times based on their information transfer to the PSD. Based on the test set, the model prediction efficiency (PE) increases with increasing L^* , ranging from -0.043 at $L^* = 3$ to 0.76 at $L^* = 6.5$. The model PE is near 0 at $L^* = 3\text{--}4$ because at this L^* range, the solar wind and magnetospheric parameters transfer little information to the PSD. Using solar wind observations at L1 and magnetospheric index (AL and SYM-H) models solely driven by solar wind, the radiation belt model can be used to forecast PSD 30–60 min ahead. This baseline model can potentially complement a class of empirical models that input data from low earth orbit (LEO).

Scientific discovery in
RB dynamics through
causality analysis

RB Neural Network
Model Assisted with
Information Theory